

EDA USING POWER BI

REPORT

WEEK – 2

**DATA MODELING & DASHBOARD CREATION - SUPERSTORE
DATASET**

SUBMITTED BY

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1. Dataset:

Superstore Sales Dataset (Preprocessed from Week 1)

2. Problem Statement:

Management wants an interactive dashboard to monitor sales, profit, orders, customers, and regional performance.

3. Objective:

To create a Star Schema data model and an interactive dashboard for Superstore sales analysis using Power BI.

4. Tools Used:

Power BI Desktop, Sample Superstore Dataset

5. Steps Performed:

Step 1: Data Cleaning & Dimensional Modeling

Created 4 Dimension Tables from the main table:

- Dim_Product (Product ID, Category, Sub-Category, Product Name)
- Dim_Customer (Customer ID, Customer Name, Segment)
- Dim_Region (Region, State, City) - Removed duplicates, kept only Region column for relationship
- Dim_Date (Order Date, Year, Month, Month Name)

Main Fact Table: samplesuperstore.

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Query: Table.Distinct("#Removed Columns", {"Product Name"})

Product ID	Category	Sub-Category	Product Name
1	OFF-PA-10000174	Office Supplies	Paper
2	OFF-BI-10004094	Office Supplies	Binders
3	OFF-LA-10003223	Office Supplies	Labels
4	OFF-ST-10002743	Office Supplies	Storage
5	OFF-AR-10003478	Office Supplies	Art
6	FUR-CH-10004063	Furniture	Chairs
7	OFF-AR-10001662	Office Supplies	Art
8	OFF-AR-10002399	Office Supplies	Art
9	OFF-BI-10004632	Office Supplies	Binders
10	OFF-FA-10001883	Office Supplies	Fasteners
11	OFF-PA-10000955	Office Supplies	Paper
12	OFF-PA-10002005	Office Supplies	Paper
13	TEC-PH-10004539	Technology	Phones
14	TEC-PH-10004977	Technology	Phones
15	FUR-FU-10004864	Furniture	Furnishings
16	OFF-BI-10003708	Office Supplies	Binders
17	OFF-AR-10004078	Office Supplies	Art
18	TEC-AC-10001266	Technology	Accessories
19	FUR-FU-10000965	Furniture	Furnishings
20	OFF-LA-10004272	Office Supplies	Labels
21	FUR-FU-10000010	Furniture	Furnishings
22	FUR-BO-10003034	Furniture	Bookcases
23	FUR-CH-10000422	Furniture	Chairs
24	OFF-AR-10003514	Office Supplies	Art

In this above screenshots, Dim_Product table has been created .

Same as Dim_Product, other Dimension tables have also been created.

Here, we can see that in the below screenshot.

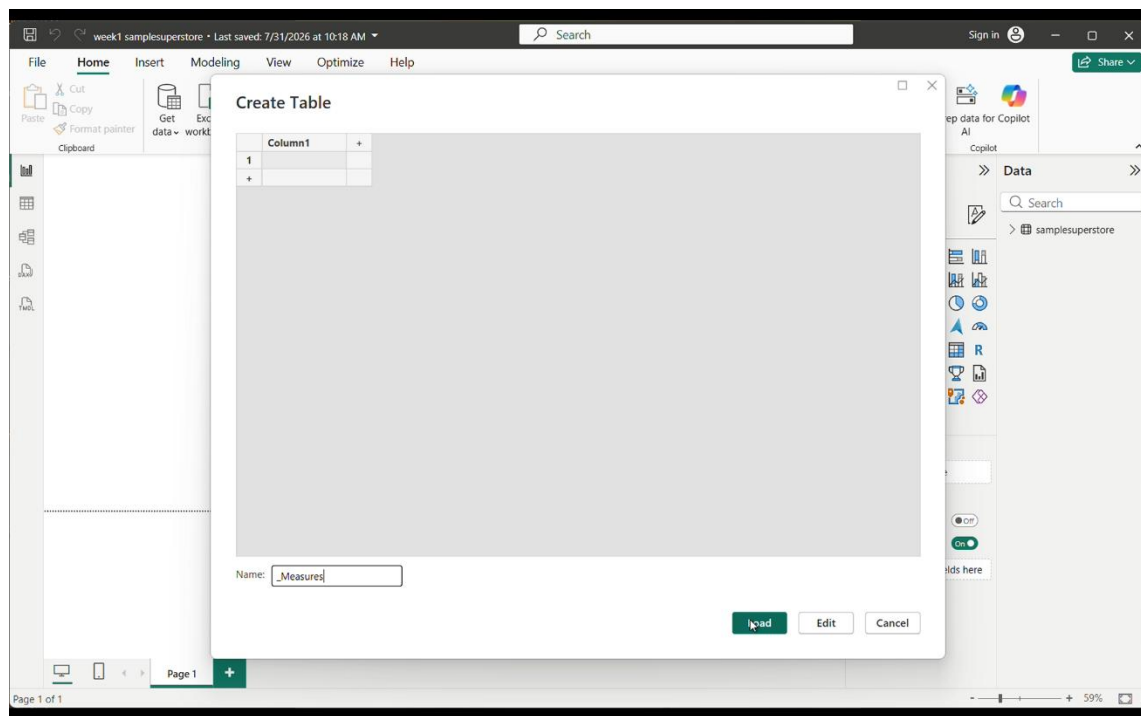
Query: Table.AddColumn("#Inserted Quarter", "Day", each Date.Day([Order Date]), Int64.Type)

Order Date	Year	Month	Quarter	Day
1	1/3/2023	2023	1	3
2	1/4/2023	2023	1	4
3	1/5/2023	2023	1	5
4	1/6/2023	2023	1	6
5	1/7/2023	2023	1	7
6	1/9/2023	2023	1	9
7	1/10/2023	2023	1	10
8	1/11/2023	2023	1	11
9	1/13/2023	2023	1	13
10	1/14/2023	2023	1	14
11	1/15/2023	2023	1	15
12	1/16/2023	2023	1	16
13	1/18/2023	2023	1	18
14	1/19/2023	2023	1	19
15	1/20/2023	2023	1	20
16	1/21/2023	2023	1	21
17	1/23/2023	2023	1	23
18	1/26/2023	2023	1	26
19	1/27/2023	2023	1	27
20	1/28/2023	2023	1	28
21	1/30/2023	2023	1	30
22	1/31/2023	2023	1	31
23	2/1/2023	2023	2	1
24	2/2/2023	2023	2	2

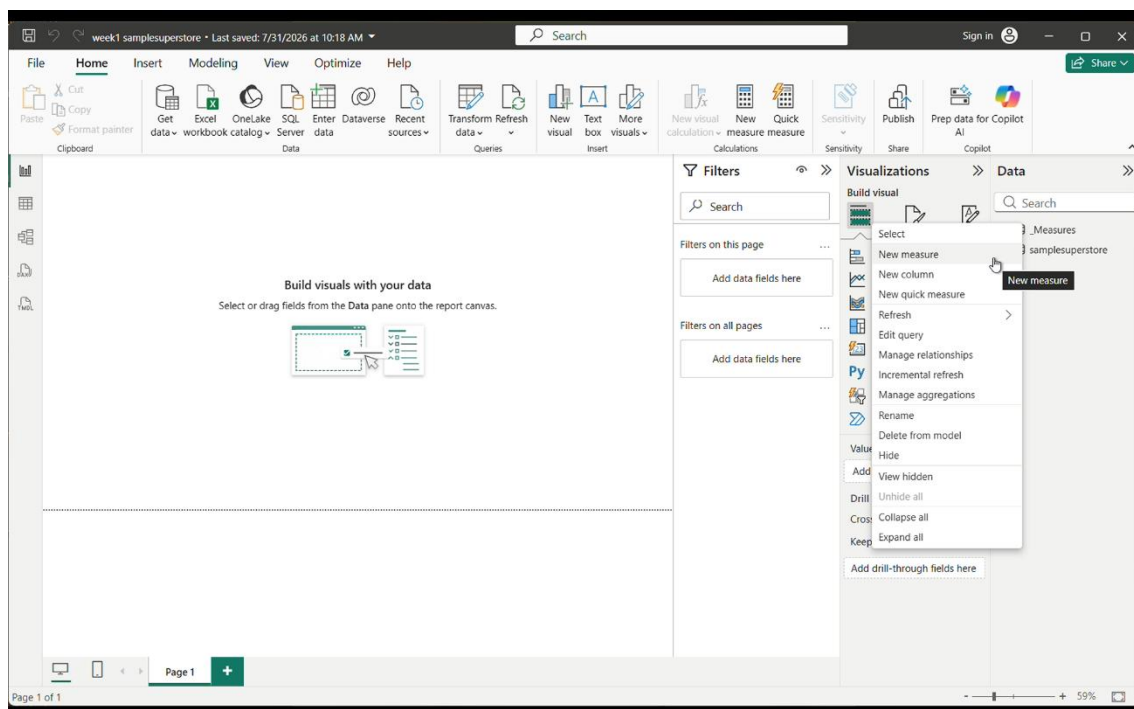
Step 2: Created Measures Table (_Measures)

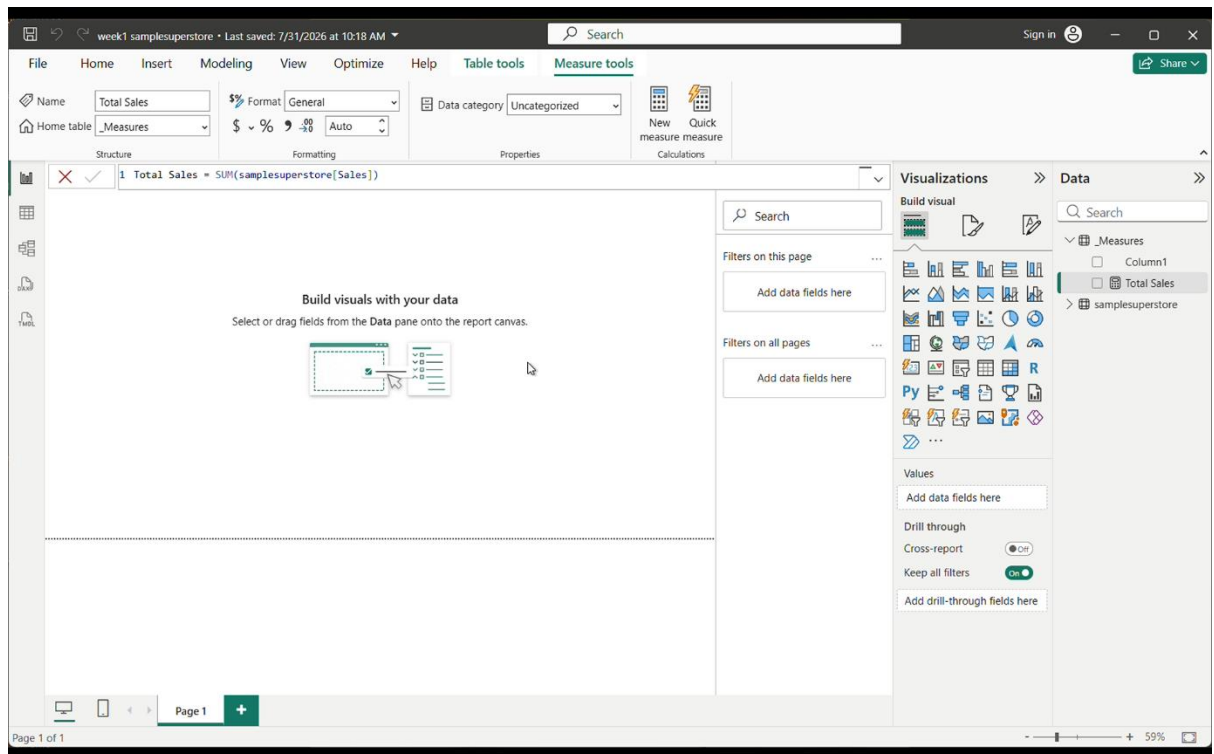
Created a separate measure table with 4 DAX measures:

- Total Sales = SUM(samplesuperstore[Sales])
- Total Profit = SUM(samplesuperstore[Profit])
- Total Orders = COUNTROWS(samplesuperstore)
- Profit Margin % = DIVIDE([Total Profit], [Total Sales])

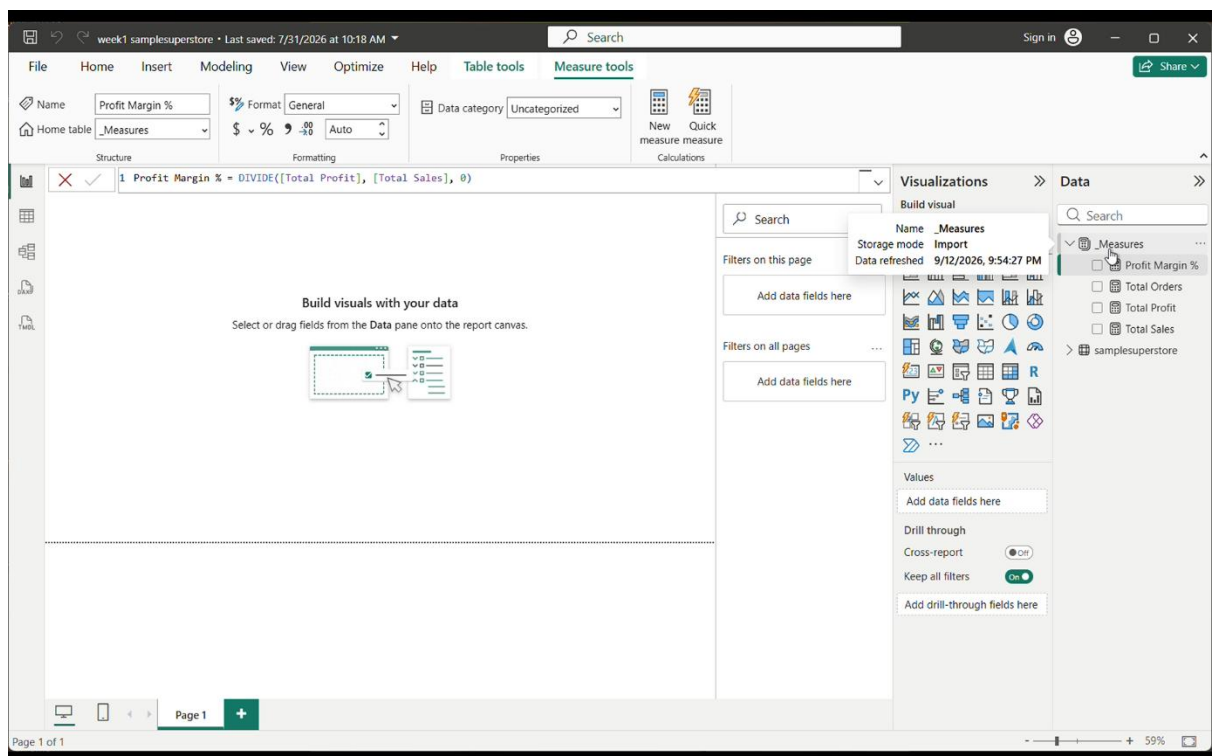


_Measures table has been created in the above screenshot.





By clicking New Measure, Total Sales has been created by using DAX measure.

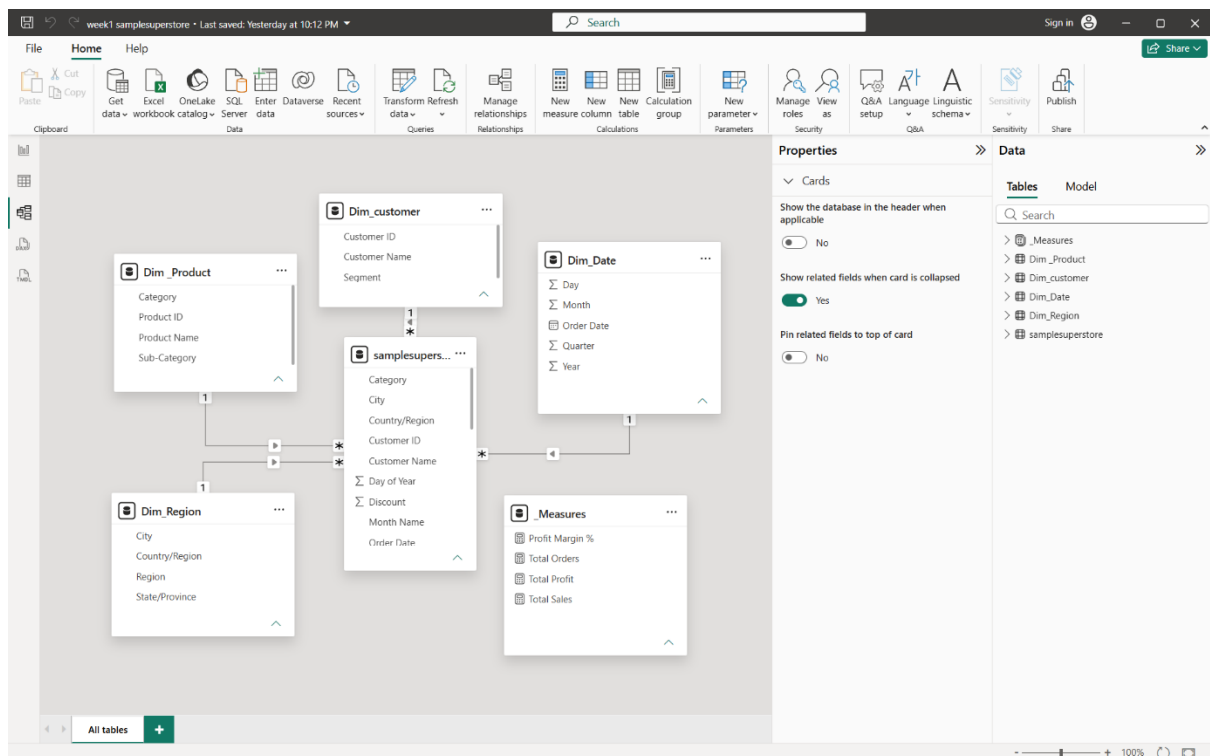


In this above screenshot we can able to see other 3 DAX Measures have also been created.

Step 3: Star Schema Modeling (Model View)

Established relationships (1 to Many, Single Direction):

- Dim_Product[Product ID] (1) -> samplesuperstore[Product ID] ()
- Dim_Customer[Customer ID] (1) -> samplesuperstore[Customer ID] ()
- Dim_Date[Order Date] (1) -> samplesuperstore[Order Date] ()
- Dim_Region (1) -> samplesuperstore () [Region]



Step 4: Dashboard Creation (Report View)*

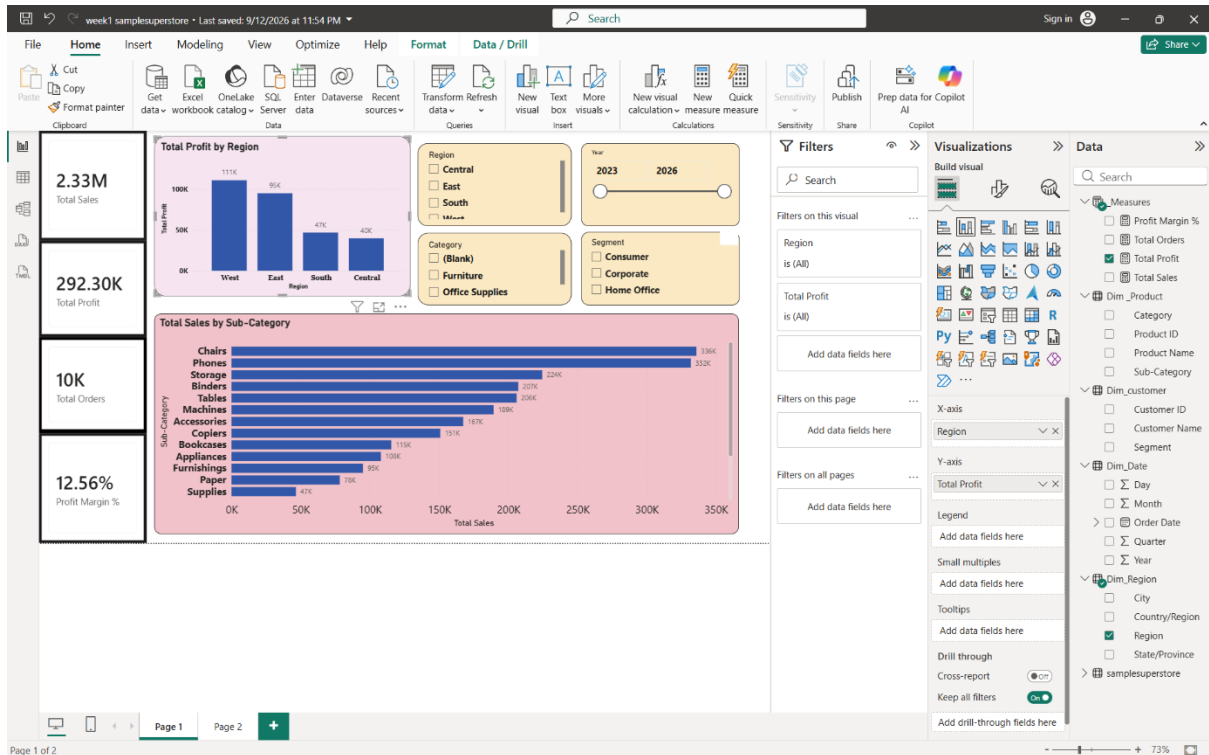
KPI Cards:

Total Sales (2.33M), Total Profit (292.30K), Total Orders (10K), Profit Margin % (12.56%)
using _Measures.

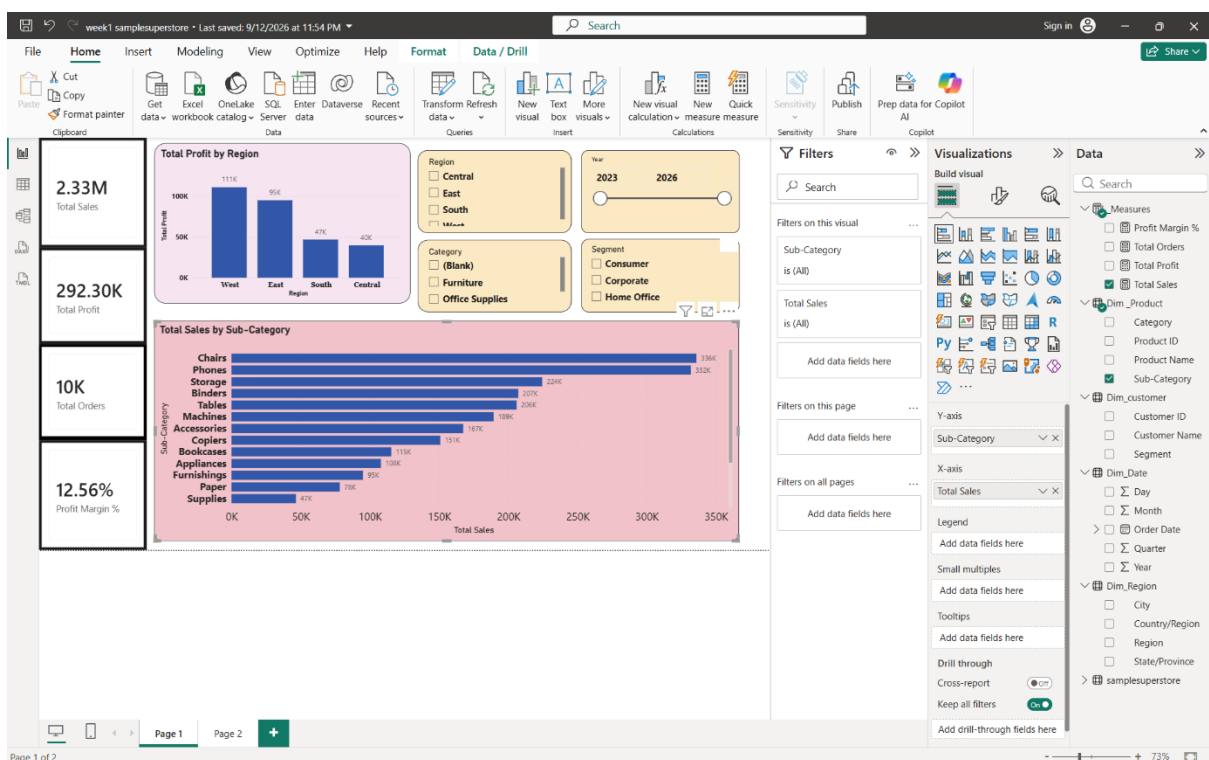
Visuals:

- Bar Chart: Total Profit by Region (X-axis: Dim_Region, Y-axis: _Measures[Total Profit])
- Bar Chart: Total Sales by Sub-Category (X-axis: Dim_Product[Sub-Category], Y-axis: _Measures[Total Sales])
- Slicers: Region (Dim_Region), Category (Dim_Product), Year (Dim_Date), Segment (Dim_Customer)[Region]

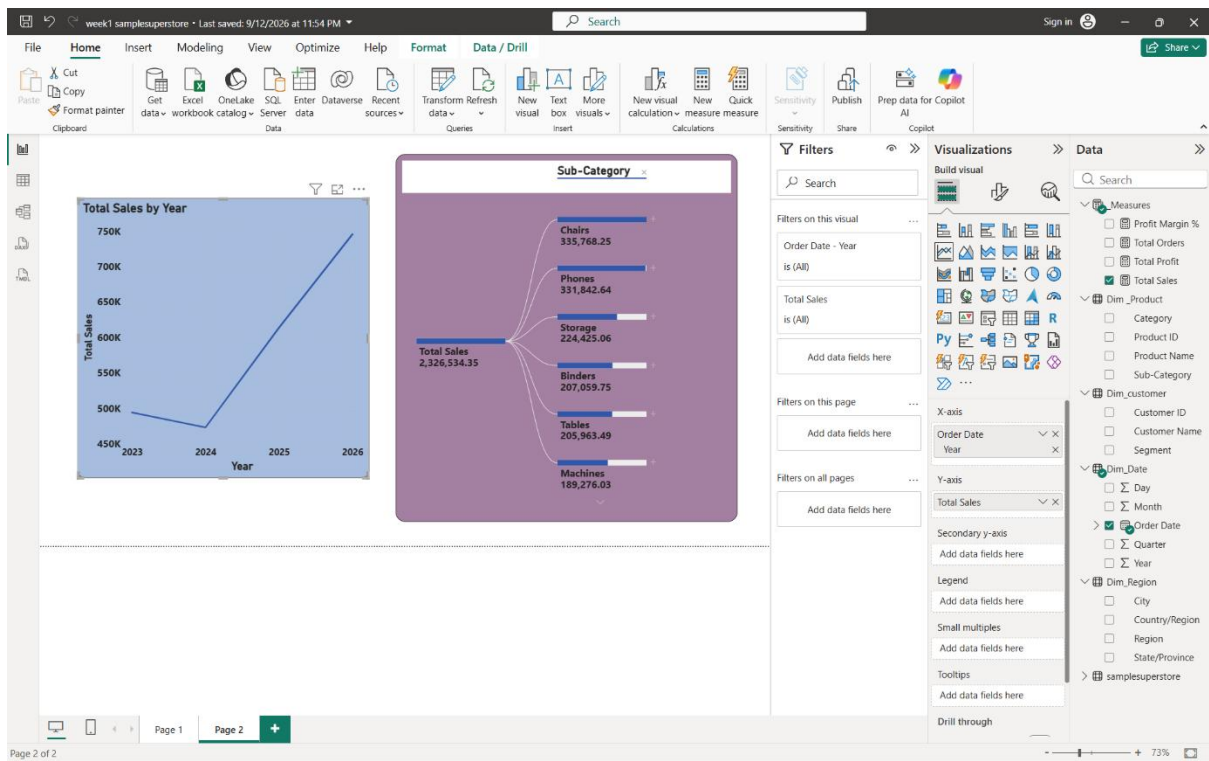
- Line Chart: Total Sales by year(X-axis: Order date -year (Dim_Date),Y-axis: Measures[Total Sales])
- Decomposition Tree: Analyse(_Measures[Total Sales]),Explain by(Dim_Product [Sub-Category,Category]),(Dim_Region[Region]).



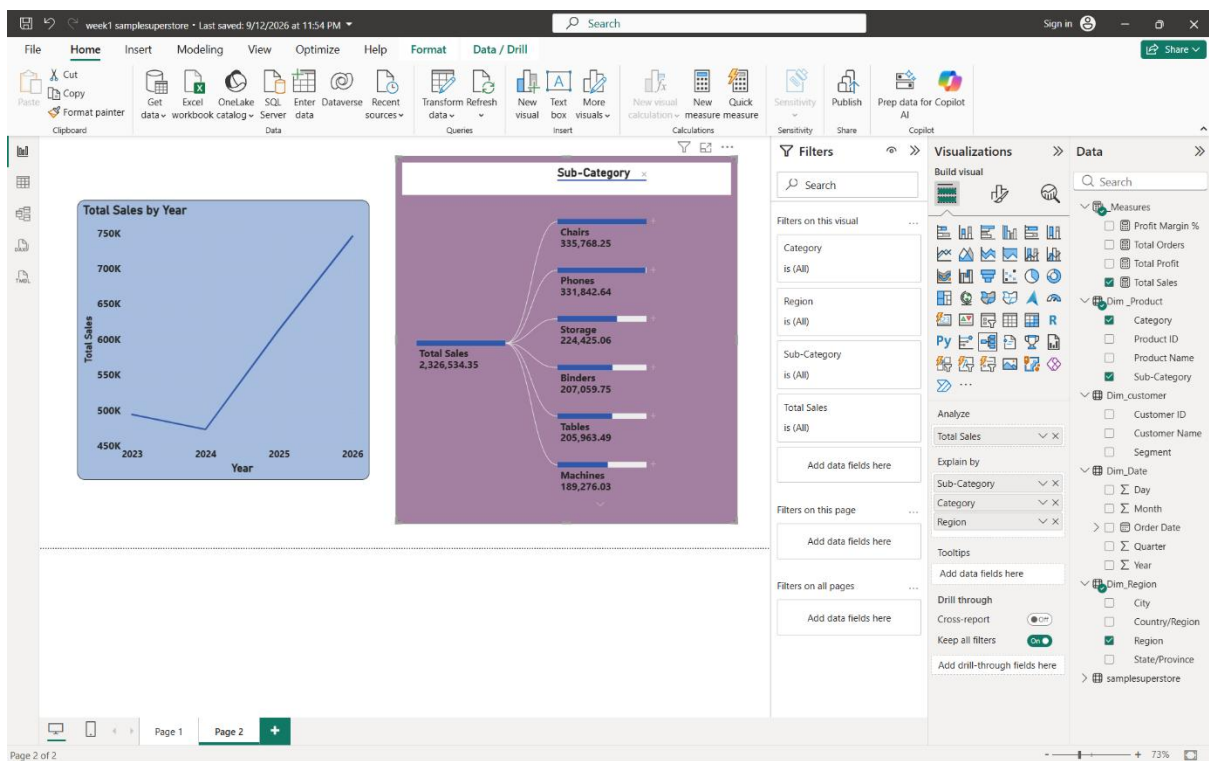
KPI cards , slicers and Stacked Column chart are shown in the above screenshot.



A Stacked Bar chart is shown in the above screenshot.



A Line chart is shown in the above screenshot.



A Decomposition Tree is shown in the above screenshot.

6. Conclusion :

Learned the importance of Star Schema for performance and accurate filtering. Understood that Measures provide the calculations (What to show) and Dimension tables provide the context (By What to filter). The dashboard now filters correctly without duplicate issues.